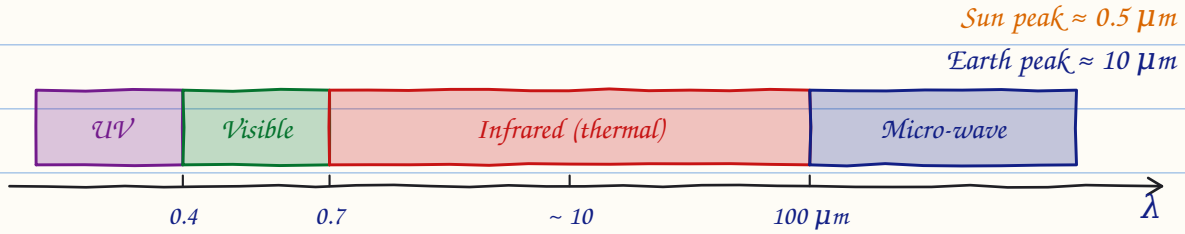
UNIT V : Radiation & Mass Transfer

Topics: thermal radiation, laws, black & gray bodies, shape factor, radiation exchange, shields, greenhouse effect; mass transfer: Fick's law, equimolal & stagnant-gas diffusion, h_m

5.1 Thermal Radiation – Basics

Thermal radiation is energy emitted by matter because of its temperature, carried by electromagnetic waves (or photons). It needs no medium and travels at the speed of light – it is the only mode that works in vacuum (Sun → Earth). Thermal band : λ = 0.1 – 100 μm (UV, visible 0.4–0.7 μm, infrared).

Thermal radiation 0.1 – 100 μm



When radiation (irradiation G) falls on a body it is partly absorbed (α), reflected (ρ) and transmitted (τ):

$$\alpha + \rho + \tau = 1$$

→ Opaque body (τ = 0) : $\alpha + \rho = 1$. Black body : $\alpha = 1$, $\rho = \tau = 0$ (absorbs all, emits maximum at every T and λ). White body : $\rho = 1$. Transparent : $\tau = 1$.

→ Reflection may be specular (mirror, angle in = angle out) or diffuse (rough surface, scattered in all directions).

→ Surface radiation: for opaque solids emission and absorption are surface phenomena (within a few μm); gases and glass are volume phenomena.

Definitions

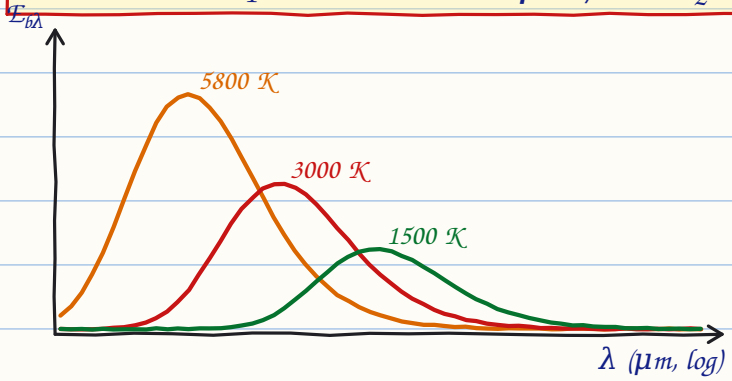
Quantity	Symbol / Unit	Meaning
Emissive power	E (W/m^2)	energy emitted per unit area, all λ, all directions
Monochromatic E_λ	$W/m^2\mu m$	emission per unit wavelength
Intensity	I ($W/m^2 sr$)	emission per unit projected area per solid angle
Irradiation G	W/m^2	radiation incident on a surface
Radiosity J	W/m^2	total radiation leaving a surface (emitted + reflected)
Emissivity ε	—	E/E_b at same T ($0 \le \varepsilon \le 1$)

Laws of Radiation

(1) Planck's law – spectral distribution of a black body

$$E_{b\lambda} = C_1 \lambda^{-5} / [\exp(C_2/\lambda T) - 1] \ (W/m^2\mu m)$$

$$C_1 = 3.742\times10^8 \ W\cdot\mu m^4/m^2, \ C_2 = 14\,388 \ \mu m\cdot K$$ Planck



- Every curve has a peak (Wien)
- Peak shifts to short λ as T rises
- Area under curve = σT^4
- Hotter curve lies above all
- (peaks not to scale)

(2) Wien's displacement law

$$\lambda_{max} T = 2898 \ \mu m\cdot K$$ Wien

→ Sun (5800 K) : λ_{max} = 2898/5800 = 0.50 μm (visible); tungsten filament 2800 K : 1.03 μm (mostly IR); human body 310 K : 9.3 μm

(3) Stefan–Boltzmann law – total emission

$$E_b = \int_0^\infty E_{b\lambda}\, d\lambda = \sigma T^4 \ \sigma = 5.67\times10^{-8} \ W/m^2K^4$$

$$\text{Real surface : } E = \varepsilon \sigma T^4$$ Stefan–Boltzmann

(4) Kirchhoff's law

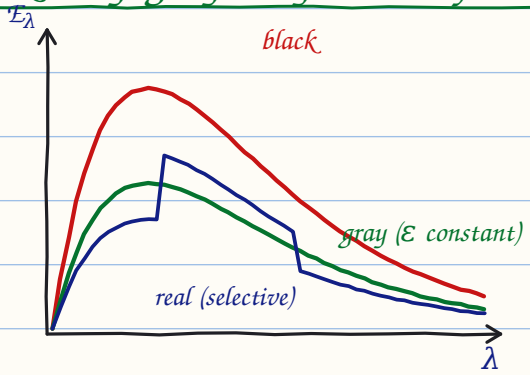
At thermal equilibrium, emissivity = absorptivity: $\varepsilon_{\lambda,\theta} = \alpha_{\lambda,\theta}$. For gray, diffuse surfaces (or when the incident radiation is from a body at the same temperature) $\varepsilon = \alpha$. A good absorber is a good emitter.

(5) Lambert's cosine law

$$I_\theta = I_n \cos\theta \text{ (emission per unit solid angle} \propto \cos\theta)$$

$$E = \pi I \text{ (diffuse surface)}$$ Lambert

Black body, gray body, real surface



Black : $\varepsilon = 1$

Gray : $\varepsilon < 1$, independent of λ

Real : ε varies with λ, T, finish

Polished Al ≈ 0.05 ; oxidised steel ≈ 0.8

Black paint 0.95 ; human skin 0.95

Net radiation from a small gray body (A, ε, T) in a large surrounding at T_{sur}: $Q = \varepsilon \sigma A (T^4 - T_{sur}^4)$

$$h_r = \varepsilon \sigma (T + T_{sur})(T^2 + T_{sur}^2) \rightarrow Q_{rad} = h_r A (T - T_{sur})$$

Solved Example 1 – Emission and Wien

A surface of 2 m^2 at 800 K, $\varepsilon = 0.8$: $Q = 0.8(5.67\times10^{-8})(2)(800)^4 = 0.8 \times 5.67\times10^{-8} \times 2 \times 4.096\times10^{11} = 37.2$ kW. $\lambda_{max} = 2898/800 = 3.62$ μm (infrared).

Solved Example 2 – Combined convection + radiation

Pipe $D = 0.1$ m, $T_s = 400$ K, $\varepsilon = 0.8$, in air and surroundings at 300 K, $h_c = 10$ W/m^2K.

$h_r = 0.8(5.67\times10^{-8})(700)(400^2+300^2) = 4.536\times10^{-8}(700)(2.5\times10^5) = 7.9$ W/m^2K

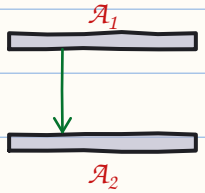
$Q/L = (h_c + h_r)\pi D (T_s - T_\infty) = (10 + 7.9)(\pi \times 0.1)(100) = 563$ W/m (radiation ≈ 44 % of the loss)

Radiation Exchange between Surfaces

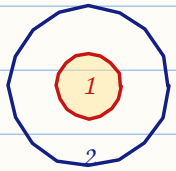
Shape (view) factor F_{12}

F_{12} = fraction of radiation leaving surface 1 (diffuse, isothermal) that directly strikes surface 2. It depends only on geometry (size, distance, orientation):

$$F_{12} = (1/A_1) \int_{A1}\int_{A2} (\cos\theta_1 \cos\theta_2 / \pi r^2) \, dA_1 \, dA_2$$



infinite parallel plates $F_{12} = 1$



enclosed body: $F_{12} = 1$, $F_{21} = A_1/A_2$

Property	Relation	Remark
Reciprocity	$A_1F_{12} = A_2F_{21}$	from symmetry of the integral
Summation	$\Sigma_j F_{ij} = 1$	enclosure of N surfaces
Plane / convex surface	$F_{ii} = 0$	cannot see itself
Concave surface	$F_{ii} > 0$	sees itself
Superposition	$F_{1(2+3)} = F_{12} + F_{13}$	add areas seen

→ Infinite long 3-surface duct (triangle, sides L_1, L_2, L_3) : $F_{12} = (L_1 + L_2 - L_3)/(2L_1)$ (Hottel crossed-string). Equilateral triangle: $F_{12} = 0.5$. For a 3-4-5 triangle, F_{12} $(L_1=3, L_2=4, L_3=5) = 2/6 = 0.33$.

→ Two coaxial parallel discs, perpendicular rectangles, etc. : read from charts (F as function of X/L, Y/L).

→ For a cylinder (1) inside concentric cylinder (2): $F_{12} = 1$, $F_{21} = r_1/r_2$, $F_{22} = 1 - r_1/r_2$.

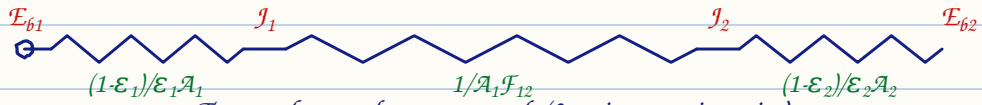
Black surfaces

$$Q_{12} = A_1F_{12}\,\sigma\,(T_1^4 - T_2^4) = (E_{b1} - E_{b2}) / R_{12}\,,\; R_{12} = 1/(A_1F_{12})$$ space resistance

Gray diffuse surfaces – radiosity–irradiation method (Oppenheim)

Radiosity $J = \varepsilon E_b + \rho G = \varepsilon E_b + (1 - \varepsilon) G$. Net heat leaving a surface : $Q = A (J - G)$ $= A \varepsilon (E_b - G)/\ldots$ which simplifies to:

$$Q_i = (E_{bi} - J_i) / [(1 - \varepsilon_i) / (\varepsilon_i A_i)] \text{ (surface resistance)}$$
$$Q_{ij} = (J_i - J_j) / [1 / (A_iF_{ij})] \text{ (space resistance)}$$



Two-surface enclosure network (3 resistances in series)

$$Q_{12} = \sigma\,(T_1^4 - T_2^4) / [(1-\varepsilon_1)/(\varepsilon_1A_1) + 1/(A_1F_{12}) + (1-\varepsilon_2)/(\varepsilon_2A_2)]$$ 2-surface enclosure

Configuration	Q_{12} (or Q/A)
Infinite parallel plates	$Q/A = \sigma(T_1^4 - T_2^4) / (1/\varepsilon_1 + 1/\varepsilon_2 - 1)$
Long concentric cylinders / spheres (1 inside)	$Q = \sigma A_1(T_1^4 - T_2^4) / [1/\varepsilon_1 + (A_1/A_2)(1/\varepsilon_2 - 1)]$
Small body in large enclosure ($A_1/A_2 \to 0$)	$Q = \varepsilon_1\,\sigma\,A_1\,(T_1^4 - T_2^4)$
Black surfaces ($\varepsilon = 1$)	$Q = \sigma\,A_1F_{12}(T_1^4 - T_2^4)$

→ 3-surface enclosures: write node equations for J_1, J_2, J_3; solve. A re-radiating (adiabatic) surface has $Q = 0 \to J = E_b$ (floating potential) – like a refractory furnace

wall.

Solved Example 3 – Concentric spheres

Sphere $D_1 = 0.2$ m, $T_1 = 500$ K, $\varepsilon_1 = 0.6$, inside a sphere $D_2 = 0.4$ m, $T_2 = 300$ K, $\varepsilon_2 = 0.8$. $A_1 = 0.1257$ m^2, $A_1/A_2 = 0.25$.

$Q = \sigma A_1(T_1^4 - T_2^4)/[1/\varepsilon_1 + (A_1/A_2)(1/\varepsilon_2 - 1)] = 5.67\times10^{-8}(0.1257)(6.25\times10^{10} - 8.1\times10^{9})/[1.667 + 0.25(0.25)]$

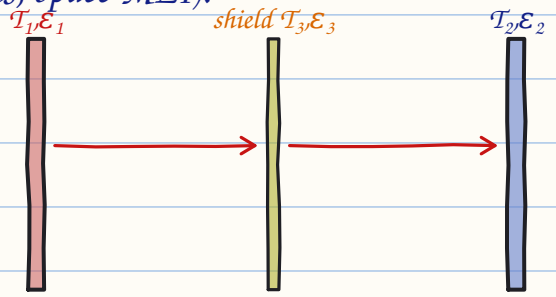
$= 387.6/1.729 =$ **224 W**

Solved Example 4 – Parallel plates

Two large parallel plates $\varepsilon_1 = \varepsilon_2 = 0.8$, $T_1 = 800$ K, $T_2 = 400$ K: $Q/A = 5.67\times10^{-8}(800^4 - 400^4)/(1/0.8 + 1/0.8 - 1) = 21\,773/1.5 =$ **14 515 W/m^2**.

Radiation Shields

A **radiation shield** is a thin, low-emissivity sheet placed between two surfaces to **increase the radiation resistance** and cut the heat flow (cryogenic tanks, thermos, thermocouple shields, space MLI).



Resistance network :

$R = (1/\varepsilon_1 - 1) + 1 + (1/\varepsilon_3 - 1) + (1/\varepsilon_3 - 1) + 1 + (1/\varepsilon_2 - 1)$

$= (1/\varepsilon_1 + 1/\varepsilon_2 - 1) + (2/\varepsilon_3 - 1)$

$$Q/A = \sigma\,(T_1^4 - T_2^4)\,/\,[\,(1/\varepsilon_1 + 1/\varepsilon_2 - 1) + \Sigma\,(2/\varepsilon_s - 1)\,]$$

If all emissivities are equal (ε) and N shields : $Q_{N\ shields} = Q_{no\ shield} / (N + 1)$

→ Shield temperature (equal emissivities, one shield): $T_3^4 = (T_1^4 + T_2^4)/2$

→ Shield material should have **very low ε** (polished Al, silver, gold-coated Mylar); more shields → smaller loss but diminishing returns and need vacuum between layers.

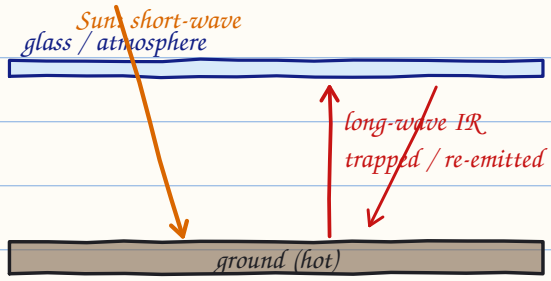
Solved Example 5 – Effect of one shield

Plates of Example 4, add one thin shield with $\varepsilon_3 = 0.05$ on both sides: $Q/A = 21\,773/[1.5 + (2/0.05 - 1)] = 21\,773/40.5 =$ **538 W/m^2** – a **96 % reduction** from 14 515.

Shield temperature from $Q/A = \sigma(T_1^4 - T_3^4)/(1/0.8 + 1/0.05 - 1)$: $T_3^4 = 4.096\times10^{11} - 538(20.25)/5.67\times10^{-8} = 2.18\times10^{11}$ → $T_3 =$ **683 K**

Greenhouse Effect

Glass, and the atmosphere, are **selectively transparent** : they transmit short-wave solar radiation (λ < 2.5–3 μm, $\tau \approx 0.9$) but are nearly **opaque to long-wave infrared** (λ > 3 μm) emitted by the warm ground (peak ≈ 10 μm). Energy comes in easily but cannot get out → the enclosure warms up.



Greenhouse gases : H_2O, CO_2, CH_4, N_2O, CFCs, O_3
Absorb at 15 μm (CO_2), 6–7 μm (H_2O) ...
Without effect : Earth ≈ -18°C
With natural effect : ≈ +15°C

Earth energy balance (solar constant S = 1361 W/m^2, albedo a = 0.30): absorbed $S(1-a)/4 = \sigma T_e^4$ → $238/5.67\times10^{-8} = 4.2\times10^9$ → T_e = 255 K (-18°C). Actual mean surface temperature ≈ 288 K (15°C); the 33 K difference is the natural greenhouse effect.

→ **Enhanced greenhouse effect** : burning fossil fuels, deforestation, agriculture raise CO_2, CH_4, N_2O → more IR absorbed → **global warming**, sea-level rise, climate change.

→ **Uses** : actual greenhouses for plants; flat-plate solar collectors (glass cover reduces re-radiation losses); car heating in sunlight.

→ **Mitigation** : renewable energy, energy efficiency, afforestation, carbon capture, reducing methane, Paris-type agreements.

Fraction of black-body emission in a wavelength band

$$F_{0-\lambda T} = \int_0^{\lambda} E_{b\lambda}\, d\lambda \;/\; \sigma T^4 \quad \text{(function of } \lambda T \text{ only)}$$

λT (μm K)	1000	2000	2898	4000	5000	6000	8000	10000
$F_{0-\lambda T}$	0.0003	0.067	0.250	0.481	0.634	0.738	0.856	0.914

Example : fraction of solar (5800 K) energy in visible band 0.4–0.7 μm : λT = 2320 → $F \approx 0.125$; λT = 4060 → $F \approx 0.49$. Fraction = 0.49 - 0.125 ≈ 0.37 (37 %). Remaining: UV ≈ 12 %, IR ≈ 51 %.

Solved Example 6 – Thermocouple radiation error

A bare thermocouple ($\varepsilon = 0.8$) in a duct reads $T_t = 500$ K. Duct wall $T_w = 400$ K, $h = 50$ W/m²K. Gas temperature T_g?

Balance : $h(T_g - T_t) = \varepsilon\sigma(T_t^4 - T_w^4) = 0.8(5.67\times10^{-8})(6.25\times10^{10} - 2.56\times10^{10}) = 1674$ W/m²

$T_g = 500 + 1674/50 = 533$ K → radiation error = 33 K. Remedy : put a polished radiation shield around the junction, raise h (suction pyrometer).

Radiation properties of common surfaces (approx. ε)

Surface	ε	Surface	ε
Polished silver	0.02	Concrete	0.88
Polished aluminium	0.05	Water (thick)	0.96
Oxidised copper	0.65	White paint	0.92
Oxidised steel	0.80	Black paint / soot	0.95–0.97

Important Questions – Radiation

1. State and explain Planck's, Wien's, Stefan–Boltzmann, Kirchhoff's and Lambert's laws.
2. Define shape factor; state its properties; derive reciprocity relation.
3. Derive net radiation exchange between two gray surfaces forming an enclosure (use network).
4. What are radiation shields? Show that for N shields (equal ε) Q reduces to 1/(N+1).
5. Explain the greenhouse effect with sketch.

Mass Transfer

Mass transfer is the movement of a chemical species from a region of higher concentration to one of lower concentration in a mixture. Examples: evaporation of water, drying of clothes, perfume spreading, absorption of gas by liquid, dissolution of sugar, humidification, membrane separation, doping of semiconductors.

→ Molecular diffusion : random molecular motion (in stagnant fluid / solids) – analogue of conduction

→ Convective mass transfer : bulk motion + diffusion between a surface and a moving

fluid – analogue of convection

Heat transfer	Mass transfer
Driving force : temperature difference	Driving force : concentration (partial pressure) difference
Fourier : $q = -k\,dT/dx$	Fick : $J = -D\,dC/dx$
Thermal diffusivity α	Mass diffusivity D_{AB}
$Nu = hL/k$, $Pr = \nu/\alpha$	$Sh = h_m L/D$, $Sc = \nu/D$
h (W/m^2K)	h_m (m/s)

Concentration measures

Quantity	Definition	Unit
Mass concentration ρ_A	mass of A / volume	kg/m^3
Molar concentration C_A	moles of A / volume $= \rho_A/M_A$	$kmol/m^3$
Mass fraction w_A	ρ_A/ρ	—
Mole fraction x_A	C_A/C	—
Partial pressure (ideal gas)	$p_A = C_A R T$, $R = 8314$ J/kmol K	Pa

→ Mixture : $\rho = \Sigma\rho_i$, $C = \Sigma C_i = P/RT$; Dalton : $P = \Sigma p_i$; $x_A = p_A/P$ for ideal gases.

Fick's law of diffusion

Mass basis : $j_A = -D_{AB}\,d\rho_A/dx$ Molar basis : $J_A = -D_{AB}\,dC_A/dx$ $(kmol/m^2s)$ Fick's first law

→ D_{AB} = diffusion coefficient (m^2/s) of A in B. Gases ≈ 10^{-5}, liquids ≈ 10^{-9}, solids 10^{-10} – 10^{-14} m^2/s. Increases with T, decreases with P (gases).

→ Gas (Gilliland) : $D_{AB} = 4.3\times10^{-3}\,T^{3/2}\,[\,1/M_A + 1/M_B\,]^{1/2} / \{\,P\,(\,V_A^{1/3} + V_B^{1/3}\,)^2\,\}$ cm^2/s (T in K, P in atm). Scaling : $D \propto T^{3/2}/P$.

→ Total molar flux in a bulk-moving mixture = diffusion + bulk flow : $N_A = -C\,D_{AB}\,dx_A/dx + x_A(N_A + N_B)$

Conservation laws & general diffusion equation

Species conservation on a control volume (rate of accumulation = net inflow + generation) gives, with constant D, C and no reaction, Fick's second law (identical to the heat conduction eqn with $\alpha \leftrightarrow D$):

$$\partial C_A/\partial t = D_{AB}\,\nabla^2 C_A \text{ (unsteady)}$$

$$\nabla^2 C_A = 0 \rightarrow d^2C_A/dx^2 = 0 \text{ (steady, 1-D, no reaction)}$$

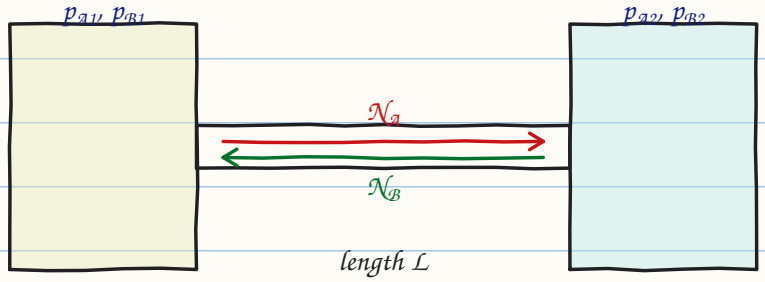
With homogeneous reaction (rate $k_1 C_A$) : $\partial C_A/\partial t = D_{AB}\nabla^2 C_A - k_1 C_A$

Overall continuity : $\partial\rho/\partial t + \nabla\cdot(\rho v) = 0$. Constitutive equations : Fick (mass), Fourier (heat), Newton (momentum) – all of the form flux = - (diffusivity) × gradient.

Steady Molecular Diffusion in Gases

(a) Isothermal equimolal counter-diffusion ($N_A = -N_B$)

Two large vessels at same T and P, joined by a tube of length L, contain mixtures A+B in different proportions. For each mole of A moving right, one mole of B moves left, so there is *no bulk flow*:



$P = p_A + p_B = const$

Linear partial-pressure profile along tube

$J_A = -J_B$

$D_{AB} = D_{BA}$

$$N_A = D_{AB}(C_{A1} - C_{A2})/L = D_{AB}(p_{A1} - p_{A2})/(RTL)\ [kmol/m^2 s]$$ equimolal

→ Since N_A is constant, $d^2C_A/dx^2 = 0$ → linear profile $C_A(x) = C_{A1} - (C_{A1} - C_{A2})\,x/L$. Mass flux = $N_A M_A$.

→ Occurs e.g. in a distillation column (binary), and in the case of "isothermal equi-mass" diffusion where molar masses are equal ($M_A = M_B$) so net mass flux is also zero.

(b) Diffusion of A through stagnant (non-diffusing) B – Stefan's law

Evaporation of liquid A at the bottom of a narrow tube into a gas stream: B is insoluble, so $N_B = 0$ and a *bulk flow* of A carries the diffusion: $N_A = J_A + x_A N_A \rightarrow N_A = -[C D_{AB}/(1 - x_A)]\, dx_A/dz$. Integrating from z_1 to z_2 :

$$N_A = [D_{AB}P/(RTz)]\ln[(P - p_{A2})/(P - p_{A1})] = [D_{AB}P/(RTz\,p_{B,lm})](p_{A1} - p_{A2})$$

$$p_{B,lm} = (p_{B2} - p_{B1})/\ln(p_{B2}/p_{B1})$$ Stefan

→ The factor $P/p_{B,lm} \geq 1$: diffusion through stagnant gas is *faster* than equimolal diffusion at same Δp because of bulk (Stefan) flow. For dilute A, $p_{B,lm} \approx P$ and the two agree.

Evaporation from a tube (liquid level falls)

Level at z (from the top). Moles evaporated : $\mathcal{N}_A = (\rho_L/M_A)\, dz/dt$. With $\mathcal{N}_A$ above and $z\, dz = [\, \mathcal{D}\, \mathcal{P}\, M_A(p_{A1} - p_{A2}) / (\rho_L\, \mathcal{R}\, \mathcal{T}\, p_{B,lm})\,]\, dt$, integrate $z_0 \rightarrow z$:

$$t = \rho_L\, \mathcal{R}\, \mathcal{T}\, p_{B,lm}\, (z^2 - z_0^2) / [\, 2\, \mathcal{D}_{AB}\, \mathcal{P}\, M_A\, (p_{A1} - p_{A2})\,]_{Winkelmann}$$

Solved Example 6 – Equimolal counter-diffusion

Tube 0.3 m long joins two tanks at 298 K, 1 atm. $\mathcal{D}_{AB} = 2\times10^{-5}\ m^2/s$; $p_{A1} = 60.8$ kPa, $p_{A2} = 20.3$ kPa.

$\mathcal{N}_A = \mathcal{D}(p_{A1} - p_{A2})/(\mathcal{R}\, \mathcal{T}\, \mathcal{L}) = 2\times10^{-5}(40\ 530)/(8314 \times 298 \times 0.3) = 0.8106/743\ 272 = 1.09\times10^{-6}\ kmol/m^2s$

Solved Example 7 – Evaporation of water in a tube

Water at 25°C evaporates into dry air at 1 atm from the bottom of a tube, $z = 0.15$ m to the open top. $p_{A1} = 3.17$ kPa, $p_{A2} = 0$, $\mathcal{D} = 2.6\times10^{-5}\ m^2/s$.

$p_{B1} = 98.16$ kPa, $p_{B2} = 101.325$ kPa; $p_{B,lm} = 3.17/\ln(101.325/98.16) = 99.7$ kPa

$\mathcal{N}_A = \mathcal{D}\, \mathcal{P}(p_{A1}-p_{A2})/(\mathcal{R}\mathcal{T}z\, p_{B,lm}) = 2.6\times10^{-5}(101.325)(3.17)/(8.314 \times 298 \times 0.15 \times 99.7)$

$= 2.25\times10^{-7}\ kmol/m^2s = 4.05\times10^{-6}\ kg/m^2s$

Diffusion in liquids and solids

→ *Liquids* : $\mathcal{D}_{AB} \approx 10^{-9}\ m^2/s$ (Wilke–Chang correlation : $\mathcal{D} = 7.4\times10^{-8}(\varphi M_B)^{1/2}T/(\mu\, V_A^{0.6})\ cm^2/s$). Concentration-dependent.

→ *Solids* : interstitial/vacancy mechanism; $\mathcal{D} = \mathcal{D}_0 \exp(-Q/\mathcal{R}T)$ (Arrhenius) e.g. carbon in steel, hydrogen in metals, drying of wood.

→ Steady diffusion through a plane membrane/wall (thickness L): $\mathcal{N}_A = \mathcal{D}_{AB}(C_{A1} - C_{A2})/L = \mathcal{D}\, S\, (p_{A1} - p_{A2})/L$ (S = solubility, Henry's law). Through a cylinder : $\mathcal{N}_A$ per unit length $= 2\pi\mathcal{D}(C_{A1}-C_{A2})/\ln(r_2/r_1)$.

Convective Mass Transfer

Mass transfer between a surface and a moving fluid is described by a *mass-transfer coefficient* h_m (analogous to h):

$$\mathcal{N}_A = h_m\, (C_{A,s} - C_{A,\infty})\ [kmol/m^2s]\ \ h_m \text{ in } m/s$$

$$\text{Mass flow } \dot{m}_A = h_m\, \mathcal{A}\, (\rho_{A,s} - \rho_{A,\infty})$$

Number	Definition	Meaning	Heat analogue
Sherwood Sh	$h_m L/D_{AB}$	convective / diffusive mass transfer	Nu
Schmidt Sc	ν/D_{AB}	momentum / mass diffusivity	Pr
Lewis Le	$\alpha/D_{AB} = Sc/Pr$	thermal / mass diffusivity	—
Stanton (mass) St_m	$h_m/V = Sh/(Re\ Sc)$		St

Correlations are obtained from the heat transfer ones by replacing $Nu \to Sh$ and $Pr \to Sc$:

Case	Correlation
Laminar flat plate (local)	$Sh_x = 0.332\ Re_x^{1/2}\ Sc^{1/3}$
Laminar flat plate (average)	$Sh_L = 0.664\ Re_L^{1/2}\ Sc^{1/3}$
Turbulent flat plate	$Sh_L = 0.037\ Re_L^{0.8}\ Sc^{1/3}$
Turbulent flow in a tube	$Sh_D = 0.023\ Re_D^{0.83}\ Sc^{1/3}$ (Gilliland–Sherwood : $Sc^{0.44}$)
Sphere (Frössling)	$Sh_D = 2 + 0.6\ Re_D^{1/2}\ Sc^{1/3}$
Free convection (vertical plate)	$Sh = C\ (Gr_m\ Sc)^n$

→ Chilton–Colburn analogy : $St\ Pr^{2/3} = St_m\ Sc^{2/3} = C_f/2 \to h/h_m = \rho\ c_p\ Le^{2/3}$ (h_m from known h, e.g. wet-bulb thermometer, cooling towers)

→ Boundary layers: $\delta/\delta_c = Sc^{1/3}$ – the concentration BL thickness plays the role of δ_t.

Solved Example 8 – Evaporation from a water surface

Air at 25°C, 2 m/s flows over a 1 m long water surface. $\nu = 1.55\times10^{-5}$, $D = 2.6\times10^{-5} \to Sc = 0.596$, $Re_L = 1.29\times10^5$ (laminar).

$Sh_L = 0.664(359)(0.596)^{1/3} = 200.7$; $h_m = Sh\ D/L = 200.7(2.6\times10^{-5})/1 = 5.2\times10^{-3}$ m/s

$C_{A,s} = p_v/RT = 3.17/(8.314 \times 298) = 1.28\times10^{-3}$ kmol/m³ ; dry air $C_{A,\infty} = 0$: $N_A = 5.2\times10^{-3}(1.28\times10^{-3}) = 6.7\times10^{-6}$ kmol/m²s ≈ 0.43 kg/m²h

Solved Example 8B – Heat–mass analogy

Air at 25°C over a surface gives $h = 20$ W/m²K. $Sc = 0.596$, $Pr = 0.70 \to Le = 0.851$; $Le^{2/3} = 0.898$. From the analogy : $h_m = h/(\rho\ c_p\ Le^{2/3}) = 20/(1.2 \times 1007 \times 0.898) =$ 0.0184 m/s.

Solved Example 9 – Diffusion through a solid wall

Hydrogen diffuses through a steel plate 5 mm thick; $C_{A1} = 2.0$ and $C_{A2} = 0.5$ kg/m^3; $D = 2.6\times10^{-13}$ m^2/s. $N_A = D(C_{A1} - C_{A2})/L = 2.6\times10^{-13}(1.5)/0.005 = 7.8\times10^{-11}$ kg/m^{2}s (very slow in solids).

Important Questions – Mass Transfer

1. State Fick's law of diffusion. Compare heat and mass transfer analogy.
2. Derive an expression for molar flux in equimolal counter diffusion.
3. Derive Stefan's expression for diffusion of A through stagnant B. Where is it used?
4. Define Sherwood, Schmidt and Lewis numbers; give a correlation for h_m.

Quick Revision

Item	Formula
Black-body emissive power	$E_b = \sigma T^4$
Wien	$\lambda_{max} T = 2898\ \mu m\cdot K$
View factor	$A_1F_{12} = A_2F_{21}$; $\Sigma F_{1j} = 1$
Two-gray-surface exchange	$\sigma\Delta(T^4)$ / [Σ resistances]
Shields (equal ε)	$Q = Q_0/(N+1)$
Fick	$N_A = -D\ dC/dx$
Equimolal	$N_A = D(p_{A1}-p_{A2})/RTL$
Stagnant B	extra factor $P/p_{B,lm}$
Sh, Sc	h_mL/D , ν/D

